SERENDIB GRAND HOTEL

Diploma in Software Engineering 24.2F

DATA MANAGEMENT 1 (Course Work – Report)

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ACKNOWLEDGEMENT

I would like to convey my deepest appreciation to all those who helped me in successfully completing this Data Management 1 coursework project, which is design and development of the database management system for Serendib Grand Hotel. This project helped me gain practical experience on entity relationship modeling, schema mapping, normalization and SQL and it would not have been possible without the guidance and encouragement provided to me throughout the process.

Most importantly, I would like to thank lecturer Mr. …… for his continued guidance, patience and constructive feedback throughout this module. He has given us an excellent explanation on concepts relating to database design and his willingness to clarify all our doubts helped us gain the necessary knowledge and confidence required to complete this project. In addition, I would like to thank the National Institute of Business Management (NIBM) for providing us all the facilities and resources necessary for this project.

In addition to this, I would like to appreciate my fellow group members for their teamwork, dedication and cooperation. Finally, I would like to thank my family and friends for their constant support and motivation for this coursework.

(K.B.D. Fernando)

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CHAPTER 1: INTRODUCTION

In the current hotel industry, an efficient operation of any hotel requires effective information management system. hotel deals with continuous flow of information each day, such as room information and price, guest information, booking information, additional services like laundry, spa and airport transfer, payments, staff information and housekeeping. When this information is recorded in paper files or spreadsheets, the problem of duplication of information, double booking of rooms, miscalculations, delay in services and report generation is created. An efficient database management system resolves these issues through providing centralized repository of all information which is relevant to the organization in organized manner. In this report, the comprehensive design and development of a database for the fictional single branch hotel named “Serendib Grand Hotel” will be discussed.

The primary purpose of the current project is to design the database which is strong, well structured and normalized for the relational model of the hotel. The database includes the room management (Standard, Deluxe and Suite rooms), guest and booking management, additional services utilization, payment processing, staff and department management and housekeeping task management. In this design, it is intended that there will be no data duplication, data integrity will be preserved and the hotel employees will be able to get access to all the required information in an efficient manner. Furthermore, the database will help in making decisions about daily activities of the hotel such as availability of rooms, housekeeping tasks which need attention, review of department-wise salaries and most importantly, the highest revenue generating rooms in the hotel. Furthermore, role-based access is provided to ensure that each employee is allowed access only to the data which he or she requires.

This report includes all the phases of the development of the database life cycle. Starting from the requirement analysis phase where entities, attributes and relations of the hotel are determined, then moving towards the conceptual design phase which includes ER diagrams, the on to the logical design phase which includes schema mapping and finally to the physical design phase which involves implementation of the database using Microsoft SQL Server. The report ends up with a brief discussion of how the database will serve the purpose of the hotel.

CHAPTER 2: ANALYSIS

Analysis involved determining the important entities, attributes and relationships needed to properly model the activities of Serendib Grand Hotel in the database. This step lays the groundwork for the subsequent design phases that follow.

Entities and Attributes Identification

A total of twelve main entities were identified which include room inventory (specialized into Standard Room, Deluxe Room and Suite), guests, staff and departments management, booking, additional services and housekeeping.

Entity	Key Attributes
Room	Room_ID, Room_Number, Floor, Price_Per_Night, Status
Standard_Room (ISA)	Room_ID, Max_Occupancy
Deluxe_Room (ISA)	Room_ID, Has_Balcony
Suite (ISA)	Room_ID, No_of_Bedrooms
Guest	Guest_ID, Guest_Name, NIC_No, Address, Phone_No (multivalued)
Booking	Booking_ID, Check_in_Date, Check_out_Date, Total_Amount
Department	Department_ID, Department_Name, Telephone_No (multivalued)
Staff	Staff_ID, Staff_Name, Age, Position, Phone_No, Salary
Manager (weak entity)	Manager_ID, Years_Worked
Service	Service_ID, Service_Name, Price
Payment	Payment_ID, Amount, Payment_Date, Payment_Method
Housekeeping_Task	Task_ID, Task_Description, Task_Date, Status

Table 1: Entities and Attributes

Defining Relationships

Understanding the relationship between different entities is critical in database designing because these relationships will define the way data in the different tables is related to one another, making sure the data is consistent. Ten relationships for the Serendib Grand Hotel have been identified, each with their respective cardinality ratios and participation constraints as indicated in Table 2 below. A staff member works in one department only and a department employs many members of staff (Works, n:1). The manager also happens to be a staff member and the manager manages one department (Has and Manage, 1:1). A guest can make many bookings and a room can also be booked many times, but each booking can be made by one guest and in one room (Places and Reserved, 1: n). There is also a many to many relationships between bookings and services (Uses) because many bookings can use

many services and a service can also be utilized by many bookings. Payments include many payments such as deposit and balance for one booking (Pays 1: n). Lastly, a Room is specialized to either Standard Room, Deluxe Room or Suite (disjoint). Participation in the column is “Total” where all instances of that entity will participate in the relationship and “Partial” where participation is not necessary.

Relationship	Between Entities	Cardinality	Participation
Works	Staff - Department	n : 1	Staff: Total, Department: Partial
Has (identifying)	Staff - Manager	1 : 1	Total
Manage	Manager - Department	1 : 1	Total (both)
Places	Guest - Booking	1 : n	Booking: Total, Guest: Partial
Reserved	Room - Booking	1 : n	Booking: Total, Room: Partial
Uses	Booking - Service	m : n	Partial (both)
Pays	Booking - Payment	1 : n	Payment: Total, Booking: Partial
Assigned	Staff - Housekeeping_Task	1 : n	Task: Total, Staff: Partial
Cleaned	Room - Housekeeping_Task	1 : n	Task: Total, Room: Partial
ISA (disjoint)	Room -> {Standard, Deluxe, Suite}	d	Total

Table 2: Relationships and Cardinality Ratios

CHAPTER 3: DESIGN

Designing the database involves putting the requirements analyzed into a database. This process is conducted in three phases. Conceptual design is the process where the entities, attributes and relationships of Serendib Grand Hotel are modeled in the form of an Entity Relationship (ER) diagram without being constrained to any database software. Logical design, on the other hand, converts this ER diagram into a relational schema using the steps in schema mapping. Subsequently, the normalization process is performed to eliminate anomalies from the schema tables created. Physical design entails implementing the schema to Microsoft SQL Server where in the tables, constraints, test data, queries and users are generated.

Conceptual Design

The Entity Relationship diagram of the system is depicted in Figure 1 using the Chen notation. Rectangles are used to represent entities while ovals are used to represent attributes with underlining being used to indicate primary keys. Double oval is used to depict multivalued attributes like the phone numbers of a Guest and telephone numbers of a department. The Manager is a weak entity shown using double rectangle since it depends on the entity Staff for its existence. The relationship between Manager and Staff is indicated using a double diamond to show that the relationship is identifying. The Room entity is specialized into Standard Room, Deluxe Room and Suite. This is indicated using a circle marked “d” where “d” indicates that the subtypes are disjoint.

The relationships will be illustrated by using diamond shapes, where the cardinality ratio (1, n or m) will be shown next to the connecting lines. Total participation is shown by a double line, where all the instances of that particular entity are required to participate in the relationship, while partial participation is illustration, all Bookings must have one to one relationship with Guest and one to many relationships with Room. Therefore, the lines of guest and room connected to booking are double. On the other hand, a Room may be available without booking it. Therefore, its line is a single one.

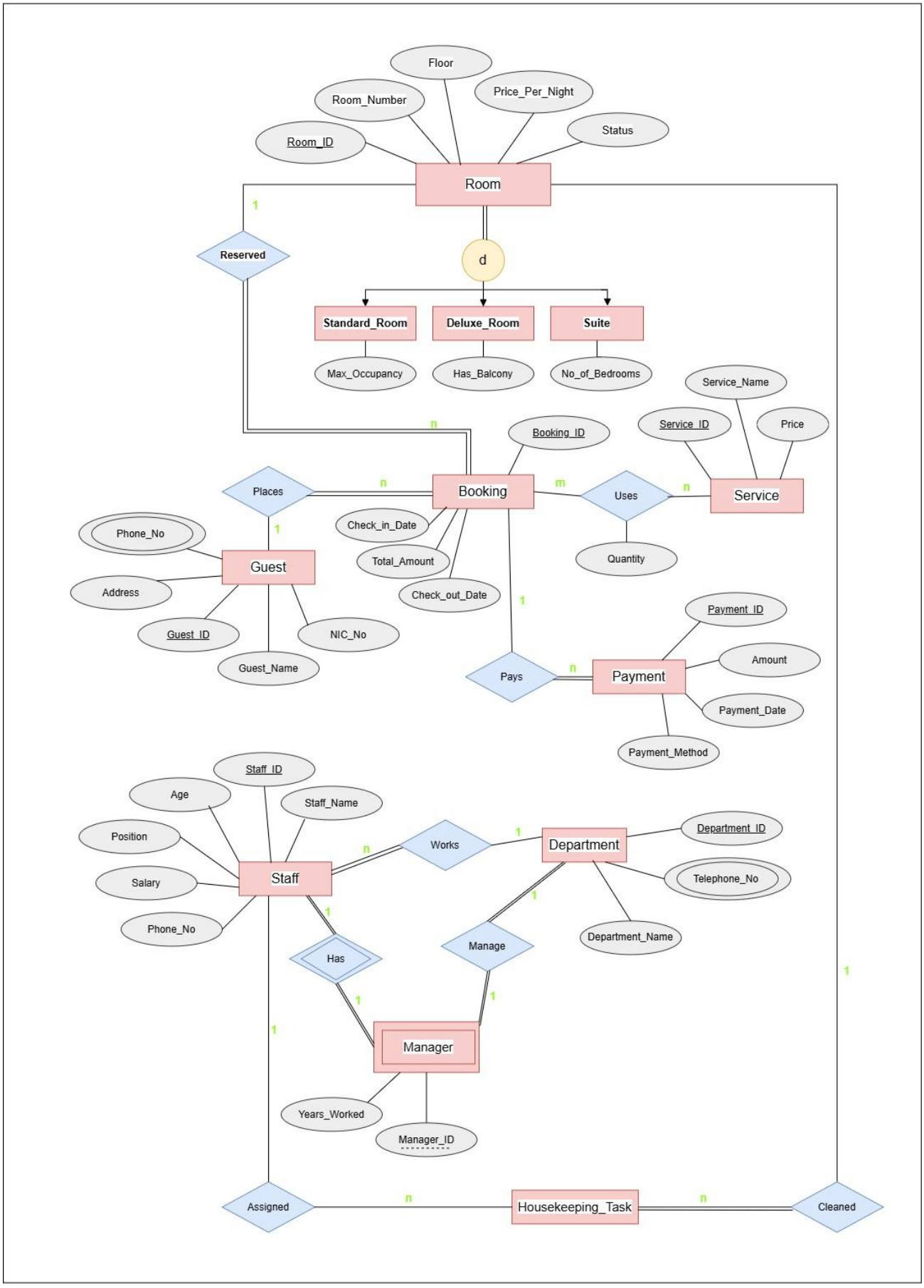

Floor
Room_Number
Price_Per_Night
Room_ID
Status
Room
1
Reserved
d
Standard_Room
Deluxe_Room
Suite
Max_Occupancy
Has_Balcony
No_of_Bedrooms
Service_Name
Service_ID
Price
n
Booking_ID
Places
n
Booking
m
Uses
n
Service
Phone_No
1
Check_in_Date
Total_Amount
Check_out_Date
Quantity
Guest
Address
Guest_ID
NIC_No
Guest_Name
1
Payment_ID
Amount
Pays
n
Payment
Payment_Date
Payment_Method
Staff_ID
Age
Staff_Name
Position
n
Works
1
Department
Department_ID
Salary
Staff
Telephone_No
Phone_No
1
1
Manage
Department_Name
Has
1
1
1
Manager
Years_Worked
Manager_ID
1
Assigned
n
Housekeeping_Task
n
Cleaned

Figure 1: Conceptual Design - ER Diagram

Logical Design

The process of logical design involves translating the ER diagram into a relational schema, which is a collection of tables which the database management system can implement without any further notification. The process of translation involves mapping all the constructs of the ER diagram, which include strong entities, weak entities, one to one relationships, one to many relationships, many to many relationships, multivalued attributes and specialization generalization ISA, Translation of the hotel ER diagram led to creation of 15 tables as seen in figure 2, since the weak entity, the disjoint room specialization, many to many service usage and two multivalued attributes needed to be represented as individual tables. After the process of schema mapping the result of the process was validated through the process of normalization.

#	Relation	Attributes (PK / FK)
1	Room	Room_ID (PK), Room_Number, Floor, Price_Per_Night, Status
2	Standard_Room	Room_ID (PK, FK), Max_Occupancy
3	Deluxe_Room	Room_ID (PK, FK), Has_Balcony
4	Suite	Room_ID (PK, FK), No_of_Bedrooms
5	Guest	Guest_ID (PK), Guest_Name, NIC_No, Address
6	Guest_Phone	Guest_ID (PK,FK), Phone_No (PK)
7	Department	Department_ID (PK), Department_Name
8	Department_Telephone	Department_ID (PK,FK), Telephone_No (PK)
9	Staff	Staff_ID (PK), Staff_Name, Age, Position, Phone_No, Salary, Department_ID (FK)
10	Manager	Manager_ID (PK), Years_Worked, Staff_ID (FK), Department_ID (FK)
11	Booking	Booking_ID (PK), Check_in_Date, Check_out_Date, Total_Amount, Guest_ID (FK), Room_ID (FK)
12	Service	Service_ID (PK), Service_Name, Price
13	Booking_Service	Booking_ID (PK,FK), Service_ID (PK,FK), Quantity
14	Payment	Payment_ID (PK), Amount, Payment_Date, Payment_Method, Booking_ID (FK)
15	Housekeeping_Task	Task_ID (PK), Task_Description, Task_Date, Status, Room_ID (FK), Staff_ID (FK)

Figure 2: Logical Design - Relational Schema

- **First Normal Form (1NF)**
  According to 1NF, an attribute should contain a single value without any repeating groups. This means that Guest and Department have been represented separately from

their multivalued attributed (Phone_number for Guest, Telephone for Department). Thus, now Guest_phone and Department_Telephone relations store each number in a separate row instead of having two (or more) phone numbers combined in a single attribute. Now each relation contains only atomic values and satisfies 1NF.

- **Second Normal Form (2NF)**
  2NF states that in a table with a composite primary key, all non-key attributes must depend on the entire key, not on parts of the key. In the Nooking_Service relation, the composite primary key (Booking_ID, Service_ID) determines the Quantity attribute. It depends on the entire key because it shows the number of units of a particular service used during a particular booking. Thus, there are no partial dependencies anywhere in the schema.

- **Third Normal Form (3NF)**
  To meet 3NF, there must be no dependencies between non-key attributes that can be dependent on the primary key via some other non-key attribute. For instance, the Staff relation has only one foreign key (Departmnet_ID) and not Department_Name, so Department_Name will reside only in the Department relation. There are no transitive dependencies because the descriptive attributes do not repeat outside their respective entities, thus, 3NF is net by all relations.

- **Boyce Codd Normal Form (BCNF)**
  To meet BCNF, all determinants should be candidate keys. For example, the determinant for the Years_Worked, Staff_ID and Department_ID attributesin Manager is Manager_ID and the determinant for the Quantity attribute in Booking_Service is the composite key of Booking_ID and Service_ID. All dependencies have a candidate key as a determinant and hence fifteen relations meet BCNF requirements.

Physical Design

The physical design stage puts into practice the logical model by implementing it as a database in Microsoft SQL Server Management Studio with the name “Serendib_Grand_Hotel”. The main emphasis of the physical design is the implementation process where tables are created with proper data types and sizes, the imposition of constraints like primary key, foreign key, Check, Unique, etc. Putting data in those tables and formulating SQL commands for accessing data.

1. Creation of Database and Tables

Firstly, the database was created and selected by using CREATE DATABASE and USE statements respectively. All the fifteen tables were created using the CREATE TABLE statement with the inclusion of the necessary primary key, foreign key, such as checkout dates that precede the check in dates. The tables were then populated with the data using the INSERT statements, having a minimum of ten record in each table.

```
SECTION 1: CREATE AND USE DATABASE
============================================================

CREATE DATABASE Serendib_Grand_Hotel;

USE Serendib_Grand_Hotel;
```

Figure 3: Physical Design - Create & Use Database

Room – Stores each room's number, floor, nightly rate and current status (Available/Occupied/Maintenance).

	Room_ID	Room_Number	Floor	Price_Per_Night	Status
1	R001	101	1	8000.00	Available
2	R002	102	1	8000.00	Available
3	R003	103	1	8500.00	Occupied
4	R004	104	1	8500.00	Available
5	R005	105	2	9000.00	Available
6	R006	106	2	9000.00	Occupied
7	R007	107	2	9200.00	Available
8	R008	108	2	9200.00	Mainten...
9	R009	109	2	9500.00	Available
10	R010	110	2	9500.00	Available
11	R011	201	3	12000.00	Available
12	R012	202	3	12000.00	Occupied
13	R013	203	3	12500.00	Available
14	R014	204	3	12500.00	Available
15	R015	205	3	13000.00	Available
16	R016	206	4	13500.00	Occupied
17	R017	207	4	14000.00	Available
18	R018	208	4	14000.00	Available
19	R019	209	4	14500.00	Available
20	R020	210	4	15000.00	Occupied
21	R021	301	5	20000.00	Available
22	R022	302	5	21000.00	Occupied
23	R023	303	5	22000.00	Available
24	R024	304	5	23000.00	Available
25	R025	305	5	24000.00	Available

Figure 4 : Physical Design - Room Table

Standard_Room/ Deluxe_Room/ Suite- Holds the attributes unique to each room category (Max_Occupancy,Has_Balcony,No_of_Bedrooms).

	Room_ID	Max_Occupancy
1	R001	2
2	R002	2
3	R003	3
4	R004	2
5	R005	2
6	R006	3
7	R007	2
8	R008	2
9	R009	3
10	R010	2

Figure 5: Physical Design - Standard Room Table

	Room_ID	Has_Balcony
1	R011	Yes
2	R012	Yes
3	R013	No
4	R014	Yes
5	R015	Yes
6	R016	No
7	R017	Yes
8	R018	Yes
9	R019	No
10	R020	Yes

Figure 6: Physical Design - Deluxe Room Table

	Room_ID	No_of_Bedrooms
1	R021	1
2	R022	2
3	R023	1
4	R024	2
5	R025	1
6	R026	2
7	R027	1
8	R028	2
9	R029	1
10	R030	2

Figure 7: Physical Design - Suite Room Table

Guest – Stores the guest's name, NIC number and Address

	Guest_ID	Guest_Name	NIC_No	Address
1	G001	Ashan Perera	891234567V	Colombo
2	G002	Nimali Fernando	925678901V	Kandy
3	G003	Ruwantha Silva	887654321V	Galle
4	G004	Dilrukshi Jayawardena	934567890V	Negombo
5	G005	Chaminda Rathnayake	199012345678	Kurunegala
6	G006	Sanduni Wickrama	200145678912	Matara
7	G007	Ishara Gunasekara	913456789V	Jaffna
8	G008	Buddhika Kariyawasam	856789012V	Anuradhapura
9	G009	Tharaka Wijesundara	199845612378	Kalutara
10	G010	Menaka Abeywardena	928765432V	Batticaloa
11	G011	Sathya Ranasinghe	200098765432	Gampaha
12	G012	Nadeeka Herath	887123456V	Ratnapura

Figure 8: Physical Design - Guest Table

Guest_Phone- Stores the guest's phone number(s) in different rows as a solution for the multivalued attribute of Phone_no using a composite key (Guest_ID, Phone_No).

	Guest_ID	Phone_No
1	G001	0711234501
2	G001	0711234502
3	G002	0712234502
4	G003	0713234503
5	G004	0714234504
6	G005	0715234505
7	G005	0715234506
8	G006	0716234506
9	G007	0717234507
10	G008	0718234508
11	G009	0719234509
12	G010	0710234510
13	G011	0711234511
14	G012	0712234512

Figure 9: Physical Design – Guest Phone Table

Department – Contains the name of each department.

	Department_ID	Department_Name
1	D01	Front Office
2	D02	Housekeeping
3	D03	Food and Beverage
4	D04	Kitchen
5	D05	Maintenance
6	D06	Security
7	D07	Human Resources
8	D08	Finance
9	D09	Sales and Marketing
10	D10	Information Technology

Figure 10: Physical Design – Department Table

Department_Telephone – Contains each department's telephone numbers.

	Department_ID	Telephone_No
1	D01	0112345001
2	D01	0112345002
3	D02	0112345010
4	D03	0112345020
5	D04	0112345030
6	D05	0112345040
7	D06	0112345050
8	D07	0112345060
9	D08	0112345070
10	D09	0112345080
11	D09	0112345081
12	D10	0112345090

Figure 11: Physical Design – Department Telephone Table

Staff – Contains each staff's name, age, designation, phone numbers, salary and references the department of each staff using the Department_ID foreign key.

	Staff_ID	Staff_Name	Age	Position	Phone_No	Salary	Department_ID
1	S001	Kasun Perera	34	Front Office Manager	0771111001	85000.00	D01
2	S002	Nadeesha Silva	28	Receptionist	0771111002	55000.00	D01
3	S003	Ruwan Fernando	40	Housekeeping Supervisor	0771111003	70000.00	D02
4	S004	Chamari Jayasinghe	30	Room Attendant	0771111004	45000.00	D02
5	S005	Dinesh Wickramasinghe	45	F and B Manager	0771111005	90000.00	D03
6	S006	Sanduni Rathnayake	26	Waiter	0771111006	40000.00	D03
7	S007	Malinda Gunawardena	38	Head Chef	0771111007	95000.00	D04
8	S008	Amaya Kumari	27	Cook	0771111008	48000.00	D04
9	S009	Nuwan Bandara	42	Maintenance Manager	0771111009	80000.00	D05
10	S010	Chathura De Silva	33	Technician	0771111010	50000.00	D05
11	S011	Priyantha Kumara	46	Security Manager	0771111011	75000.00	D06
12	S012	Lakshan Peris	29	Security Guard	0771111012	42000.00	D06
13	S013	Harshani Wijesinghe	35	HR Manager	0771111013	88000.00	D07
14	S014	Tharindu Madushanka	31	HR Executive	0771111014	55000.00	D07
15	S015	Iresha Abeysekara	39	Finance Manager	0771111015	92000.00	D08
16	S016	Roshan Dissanayake	30	Accountant	0771111016	60000.00	D08
17	S017	Sachini Pathirana	32	Sales Manager	0771111017	87000.00	D09
18	S018	Kavindu Rajapaksha	28	Marketing Executive	0771111018	52000.00	D09
19	S019	Manoj Senanayake	41	IT Manager	0771111019	95000.00	D10
20	S020	Yohan Fonseka	27	IT Support	0771111020	50000.00	D10

Figure 12: Physical Design – Staff Table

Manager - Manager is a weak entity containing years of working and references.

	Manager_ID	Years_Worked	Staff_ID	Department_ID
1	M001	5	S001	D01
2	M002	7	S003	D02
3	M003	10	S005	D03
4	M004	8	S007	D04
5	M005	6	S009	D05
6	M006	9	S011	D06
7	M007	4	S013	D07
8	M008	11	S015	D08
9	M009	3	S017	D09
10	M010	6	S019	D10

Figure 13: Physical Design – Manager Table

Booking - Contains check in/ Check out date, total amount of each reservation and references the guest who made it and the Room it is made for using the Guest-ID and Room_ID foreign keys.

	Booking_ID	Check_in_Date	Check_out_Date	Total_Amount	Guest_ID	Room_ID
1	B001	2026-08-01	2026-08-03	16000.00	G001	R001
2	B002	2026-08-02	2026-08-05	36000.00	G002	R012
3	B003	2026-08-03	2026-08-06	63000.00	G003	R022
4	B004	2026-08-04	2026-08-05	8500.00	G004	R003
5	B005	2026-08-05	2026-08-08	40500.00	G005	R016
6	B006	2026-08-06	2026-08-07	9000.00	G006	R006
7	B007	2026-08-07	2026-08-10	45000.00	G007	R020
8	B008	2026-08-08	2026-08-11	78000.00	G008	R027
9	B009	2026-08-09	2026-08-10	9200.00	G009	R008
10	B010	2026-08-10	2026-08-12	24000.00	G010	R011
11	B011	2026-08-11	2026-08-14	66000.00	G011	R023
12	B012	2026-08-12	2026-08-13	8000.00	G012	R002
13	B013	2026-09-01	2026-09-04	42000.00	G001	R017
14	B014	2026-09-02	2026-09-03	9500.00	G003	R009
15	B015	2026-09-03	2026-09-06	81000.00	G005	R029

Figure 14:Physical Design – Booking Table

Service – Contains the name and cost of each service offered and is referenced by Booking_Service.

	Service_ID	Service_Name	Price
1	SV01	Room Service	2500.00
2	SV02	Laundry	1500.00
3	SV03	Spa Treatment	6000.00
4	SV04	Airport Pickup	4000.00
5	SV05	Gym Access	1000.00
6	SV06	Mini Bar	2000.00
7	SV07	Extra Bed	3000.00
8	SV08	Late Checkout	2500.00
9	SV09	Breakfast Buffet	1800.00
10	SV10	Car Parking	500.00

Figure 15: Physical Design – Service Table

Booking_Service – Solves the many to many relationships between Booking and Service.

	Booking_ID	Service_ID	Quantity
1	B001	SV01	2
2	B001	SV09	2
3	B002	SV02	1
4	B002	SV06	1
5	B003	SV03	1
6	B003	SV04	1
7	B004	SV01	1
8	B005	SV07	1
9	B005	SV09	3
10	B006	SV06	2
11	B007	SV03	2
12	B008	SV04	1
13	B008	SV08	1
14	B010	SV09	2

Figure 16: Physical Design – Booking Service Table

Payment – Contains information about the amount, date and type of payment

	Payment_ID	Amount	Payment_Date	Payment_Method	Booking_ID
1	P001	8000.00	2026-08-01	Cash	B001
2	P002	8000.00	2026-08-03	Card	B001
3	P003	36000.00	2026-08-02	Card	B002
4	P004	30000.00	2026-08-03	Online	B003
5	P005	33000.00	2026-08-06	Card	B003
6	P006	8500.00	2026-08-04	Cash	B004
7	P007	20000.00	2026-08-05	Card	B005
8	P008	20500.00	2026-08-08	Cash	B005
9	P009	9000.00	2026-08-06	Cash	B006
10	P010	45000.00	2026-08-07	Online	B007
11	P011	40000.00	2026-08-08	Card	B008
12	P012	38000.00	2026-08-11	Cash	B008
13	P013	9200.00	2026-08-09	Card	B009
14	P014	24000.00	2026-08-10	Online	B010
15	P015	66000.00	2026-08-11	Card	B011
16	P016	8000.00	2026-08-12	Cash	B012
17	P017	42000.00	2026-09-01	Card	B013
18	P018	9500.00	2026-09-02	Cash	B014
19	P019	81000.00	2026-09-03	Online	B015

Figure 17: Physical Design – Payment Table

Housekeeping_Task – Contains information about each task.

	Task_ID	Task_Description	Task_Date	Status	Room_ID	Staff_ID
1	T001	Clean and prepare room	2026-08-01	Completed	R001	S004
2	T002	Restock mini bar	2026-08-02	Completed	R012	S004
3	T003	Deep clean bathroom	2026-08-03	Ongoing	R022	S003
4	T004	Change bedsheets	2026-08-04	Completed	R003	S004
5	T005	Fix air conditioning	2026-08-05	Completed	R016	S010
6	T006	Clean carpet	2026-08-06	Pending	R006	S004
7	T007	Repair shower	2026-08-07	Ongoing	R020	S010
8	T008	General cleaning	2026-08-08	Completed	R027	S003
9	T009	Replace light bulb	2026-08-09	Completed	R008	S010
10	T010	Vacuum room	2026-08-10	Pending	R011	S004
11	T011	Clean pool view suite	2026-08-11	Ongoing	R023	S003
12	T012	Inspect minibar	2026-08-12	Completed	R002	S004

Figure 18: Physical Design – Housekeeping Task Table

2. SQL SELECT Queries

Five frequent data retrieval requirements have been specified and realized through SELECT queries, which include the following: available rooms, highly paid employees, future booking, pending cleaning assignments and guest information

along with their telephone number. The use of the WHERE statement to select specific rows and the implicit JOIN of the final example are illustrated by these queries.

```sql
-- Q1: Display all rooms that are currently Available
SELECT Room_ID, Room_Number, Floor, Price_Per_Night
FROM Room
WHERE Status = 'Available';


-- Q2: Display staff earning a salary greater than 80000
SELECT Staff_Name, Position, Salary
FROM Staff
WHERE Salary > 80000;


-- Q3: Display upcoming bookings (check-in date after 2026-08-31)
SELECT Booking_ID, Check_in_Date, Check_out_Date, Total_Amount
FROM Booking
WHERE Check_in_Date > '2026-08-31';


-- Q4: Display housekeeping tasks that are still Pending
SELECT Task_ID, Task_Description, Task_Date, Room_ID
FROM Housekeeping_Task
WHERE Status = 'Pending';


-- Q5: Display guest details combined with their phone numbers
SELECT Guest.Guest_ID, Guest.Guest_Name, Guest.Address, Guest_Phone.Phone_No
FROM Guest, Guest_Phone
WHERE Guest.Guest_ID = Guest_Phone.Guest_ID;
```

Figure 19: Physical Design – SQL SELECT Queries

Question 1 – Filter Room table where Status is ‘Available’ to list all available rooms with floor and cost per night.

	Room_ID	Room_Number	Floor	Price_Per_Night
1	R001	101	1	8000.00
2	R002	102	1	8000.00
3	R004	104	1	8500.00
4	R005	105	2	9000.00
5	R007	107	2	9200.00
6	R009	109	2	9500.00
7	R010	110	2	9500.00
8	R011	201	3	12000.00
9	R013	203	3	12500.00
10	R014	204	3	12500.00
11	R015	205	3	13000.00
12	R017	207	4	14000.00
13	R018	208	4	14000.00
14	R019	209	4	14500.00
15	R021	301	5	20000.00
16	R023	303	5	22000.00
17	R024	304	5	23000.00
18	R025	305	5	24000.00
19	R026	306	5	25000.00
20	R028	402	6	26000.00

Figure 20: Physical Design – Room by Status = 'Available'

Question 2 – Filter Staff table where Salary is greater than 80000 to list high paid employees.

	Staff_Name	Position	Salary
1	Kasun Perera	Front Office Manager	85000.00
2	Dinesh Wickramasinghe	F and B Manager	90000.00
3	Malinda Gunawardena	Head Chef	95000.00
4	Harshani Wijesinghe	HR Manager	88000.00
5	Iresha Abeysekara	Finance Manager	92000.00
6	Sachini Pathirana	Sales Manager	87000.00
7	Manoj Senanayake	IT Manager	95000.00

Figure 21: Physical Design – Staff by Salary > 80000

Question 3 – Filter Booking table where Check_in_Date is later than ‘2026-08-31’ to list future bookings.

	Booking_ID	Check_in_Date	Check_out_Date	Total_Amount
1	B013	2026-09-01	2026-09-04	42000.00
2	B014	2026-09-02	2026-09-03	9500.00
3	B015	2026-09-03	2026-09-06	81000.00

Figure 22: Physical Design – Booking by Check in Date > '2026-08-31'

Question 4 – Filter Housekeeping_Task table where Status is ‘Pending’ to list pending tasks of maintenance.

	Task_ID	Task_Description	Task_Date	Room_ID
1	T006	Clean carpet	2026-08-06	R006
2	T010	Vacuum room	2026-08-10	R011

Figure 23: Physical Design – Housekeeping Task by Status = 'Pending'

Question 5 – Join Guest and Guest_Phone tables where Guest_ID is Same.

	Guest_ID	Guest_Name	Address	Phone_No
1	G001	Ashan Perera	Colombo	0711234501
2	G001	Ashan Perera	Colombo	0711234502
3	G002	Nimali Fernando	Kandy	0712234502
4	G003	Ruwantha Silva	Galle	0713234503
5	G004	Dilrukshi Jayawardena	Negombo	0714234504
6	G005	Chaminda Rathnayake	Kurunegala	0715234505
7	G005	Chaminda Rathnayake	Kurunegala	0715234506
8	G006	Sanduni Wickrama	Matara	0716234506
9	G007	Ishara Gunasekara	Jaffna	0717234507
10	G008	Buddhika Kariyawasam	Anuradhapura	0718234508
11	G009	Tharaka Wijesundara	Kalutara	0719234509
12	G010	Menaka Abeywardena	Batticaloa	0710234510
13	G011	Sathya Ranasinghe	Gampaha	0711234511
14	G012	Nadeeka Herath	Ratnapura	0712234512

Figure 24: Physical Design – Joins Guest and Guest Phone on Guest ID

3. Advanced SQL Queries

Five advanced SQL queries were developed by utilizing GROUP BY, HAVING and ORDER BY to respond to the inquiries that need aggregation in addition to filtering, for instance, the number of employees in each department, total cost of salaries for each department, departments above the salary limit, most income producing rooms and guest who have made more than one booking. The difference between WHERE and HAVING is demonstrated through these queries.

```sql
-- A1: Number of staff working in each department
SELECT Department.Department_Name, COUNT(Staff.Staff_ID) AS Number_Of_Staff
FROM Department, Staff
WHERE Department.Department_ID = Staff.Department_ID
GROUP BY Department.Department_Name;


-- A2: Total salary paid in each department
SELECT Department.Department_Name, SUM(Staff.Salary) AS Total_Salary
FROM Department, Staff
WHERE Department.Department_ID = Staff.Department_ID
GROUP BY Department.Department_Name;


-- A3: Departments with a total salary greater than 130000, ordered descending
SELECT Department.Department_Name, SUM(Staff.Salary) AS Total_Salary
FROM Department, Staff
WHERE Department.Department_ID = Staff.Department_ID
GROUP BY Department.Department_Name
HAVING SUM(Staff.Salary) > 130000
ORDER BY SUM(Staff.Salary) DESC;
```

```sql
-- A4: Total revenue generated by each room, ordered by revenue descending
SELECT Room_ID, SUM(Total_Amount) AS Total_Revenue
FROM Booking
GROUP BY Room_ID
ORDER BY Total_Revenue DESC;


-- A5: Guests who have made more than one booking
SELECT Guest_ID, COUNT(Booking_ID) AS Number_Of_Bookings
FROM Booking
GROUP BY Guest_ID
HAVING COUNT(Booking_ID) > 1;
```

Figure 25: Physical Design – Advanced SQL Queries

A1 – connects Department and Staff tables, GROUPS BY Department_Name and COUNT (), displays how many people work in each department.

	Department_Name	Number_Of_Staff
1	Finance	2
2	Food and Beverage	2
3	Front Office	2
4	Housekeeping	2
5	Human Resources	2
6	Information Technology	2
7	Kitchen	2
8	Maintenance	2
9	Sales and Marketing	2
10	Security	2

Figure 26: Physical Design – GROUPS BY Department Name and COUNT ()

A2 – connects Department and Staff tables, GROUPS BY Department_Name and uses SUM () to show the salary cost of each department.

	Department_Name	Total_Salary
1	Finance	152000.00
2	Food and Beverage	130000.00
3	Front Office	140000.00
4	Housekeeping	115000.00
5	Human Resources	143000.00
6	Information Technology	145000.00
7	Kitchen	143000.00
8	Maintenance	130000.00
9	Sales and Marketing	139000.00
10	Security	117000.00

Figure 27: Physical Design – GROUPS BY Department Name and uses SUM ()

A3 – extends A2 using HAVING SUM(Salary) > 130000 and ORDER BY DESC, filters only expensive departments, sorted by their costs from the highest to the lowest.

	Department_Name	Total_Salary
1	Finance	152000.00
2	Information Technology	145000.00
3	Kitchen	143000.00
4	Human Resources	143000.00
5	Front Office	140000.00
6	Sales and Marketing	139000.00

Figure 28: Physical Design – HAVING SUM(Salary) > 130000 and ORDER BY DESC

A4 – GROUPS Booking table by Room_ID and uses SUM(Total_Amount) and ORDER BY DESC, finds rooms that generate the highest amount of income.

	Room_ID	Total_Revenue
1	R029	81000.00
2	R027	78000.00
3	R023	66000.00
4	R022	63000.00
5	R020	45000.00
6	R017	42000.00
7	R016	40500.00
8	R012	36000.00
9	R011	24000.00
10	R001	16000.00
11	R009	9500.00
12	R008	9200.00
13	R006	9000.00
14	R003	8500.00
15	R002	8000.00

Figure 29: Physical Design – GROUPS by and ORDER BY DESC

A5 – GROUPS Booking table by Guest_ID and uses COUNT () and HAVING COUNT () > 1, filters guests that have booked a room more than once.

	Guest_ID	Number_Of_Bookings
1	G001	2
2	G003	2
3	G005	2

Figure 30: Physical Design – GROUPS by and HAVING COUNT () > 1

4. Joining Tables

There were three INNER JOIN statements used to join information that was distributed in various but related tables such as joining Department, Manager and Staff to find the manager of each department; Guest, Booking and Room to display the booking information; and Booking, Booking_Service and Service to display additional service in each booking.

```sql
-- J1: Join Department, Manager and Staff to show each department's manager
SELECT Department.Department_Name, Staff.Staff_Name AS Manager_Name, Manager.Years_Worked
FROM (Department INNER JOIN Manager ON Department.Department_ID = Manager.Department_ID)
INNER JOIN Staff ON Manager.Staff_ID = Staff.Staff_ID
ORDER BY Department.Department_Name;

-- J2: Join Guest, Booking and Room to show who booked which room and when
SELECT Guest.Guest_Name, Room.Room_Number, Booking.Check_in_Date, Booking.Check_out_Date
FROM (Guest INNER JOIN Booking ON Guest.Guest_ID = Booking.Guest_ID)
INNER JOIN Room ON Booking.Room_ID = Room.Room_ID
ORDER BY Booking.Check_in_Date;

-- J3: Join Booking, Booking_Service and Service to show services used per booking
SELECT Booking.Booking_ID, Service.Service_Name, Booking_Service.Quantity
FROM (Booking INNER JOIN Booking_Service ON Booking.Booking_ID = Booking_Service.Booking_ID)
INNER JOIN Service ON Booking_Service.Service_ID = Service.Service_ID
ORDER BY Booking.Booking_ID;
```

Figure 31: Physical Design – Tables Join Queries

J1 – Inner join of Departments, Managers and Staff tables to show which staff member manages which department and their experience.

	Department_Name	Manager_Name	Years_Worked
1	Finance	Iresha Abeysekara	11
2	Food and Beverage	Dinesh Wickramasinghe	10
3	Front Office	Kasun Perera	5
4	Housekeeping	Ruwan Fernando	7
5	Human Resources	Harshani Wijesinghe	4
6	Information Technology	Manoj Senanayake	6
7	Kitchen	Malinda Gunawardena	8
8	Maintenance	Nuwan Bandara	6
9	Sales and Marketing	Sachini Pathirana	3
10	Security	Priyantha Kumara	9

Figure 32: Physical Design – Join of Departments, Managers and Staff tables

J2 – Inner join of Guests, Bookings and Rooms tables to show which guest has made a booking for which room with check-in and check-out dates.

	Guest_Name	Room_Number	Check_in_Date	Check_out_Date
1	Ashan Perera	101	2026-08-01	2026-08-03
2	Nimali Fernando	202	2026-08-02	2026-08-05
3	Ruwantha Silva	302	2026-08-03	2026-08-06
4	Dilrukshi Jayawardena	103	2026-08-04	2026-08-05
5	Chaminda Rathnayake	206	2026-08-05	2026-08-08
6	Sanduni Wickrama	106	2026-08-06	2026-08-07
7	Ishara Gunasekara	210	2026-08-07	2026-08-10
8	Buddhika Kariyawasam	401	2026-08-08	2026-08-11
9	Tharaka Wijesundara	108	2026-08-09	2026-08-10
10	Menaka Abeywardena	201	2026-08-10	2026-08-12
11	Sathya Ranasinghe	303	2026-08-11	2026-08-14
12	Nadeeka Herath	102	2026-08-12	2026-08-13
13	Ashan Perera	207	2026-09-01	2026-09-04
14	Ruwantha Silva	109	2026-09-02	2026-09-03
15	Chaminda Rathnayake	403	2026-09-03	2026-09-06

Figure 33: Physical Design – Join of Guests, Bookings and Rooms tables

J3 – Inner join of Bookings, Booking_Service and Services tables to show which extra services and their quantities have been used.

	Booking_ID	Service_Name	Quantity
1	B001	Room Service	2
2	B001	Breakfast Buffet	2
3	B002	Laundry	1
4	B002	Mini Bar	1
5	B003	Spa Treatment	1
6	B003	Airport Pickup	1
7	B004	Room Service	1
8	B005	Extra Bed	1
9	B005	Breakfast Buffet	3
10	B006	Mini Bar	2
11	B007	Spa Treatment	2
12	B008	Airport Pickup	1
13	B008	Late Checkout	1
14	B010	Breakfast Buffet	2

Figure 34: Physical Design – Join of Bookings, Booking Service and Services tables

5. User-Level Permissions

Seven users have been assigned roles by the execution of CREATE USER and GRANT statements on the basis of the principle of least privilege. Ownership and management of the database have been granted the privilege of CONTROL, whereas the managers for each department (HR, Front Office, Housekeeping, and Finance) have been assigned ALL permissions on the relevant tables only, while the Receptionist has been allowed SELECT permission to check availability.

- Vajira (Owner): Providing all privileges in the database to the hotel's owner.

```
-- 1) Owner of the hotel - full control over the entire database
CREATE USER Vajira FOR LOGIN Owner;
GRANT CONTROL TO Vajira;
```

Figure 35: Physical Design – Permission to Owner

- Manoj (IT Manager): Providing all privileges in the database to the IT Manager responsible for creating and managing the database.

```
-- 2) IT Manager who created and maintains the database - full control
CREATE USER Manoj FOR LOGIN ITManager;
GRANT CONTROL TO Manoj;
```

Figure 36: Physical Design – Permission to IT Manager

- Harshani (HR Manager): Providing full privileges to staff and department-related tables for HR purposes.

```
-- 3) HR Manager - full access limited to HR-related tables only
CREATE USER Harshani FOR LOGIN HRManager;
GRANT ALL ON Staff TO Harshani;
GRANT ALL ON Department TO Harshani;
GRANT ALL ON Department_Telephone TO Harshani;
GRANT ALL ON Manager TO Harshani;
```

Figure 37: Physical Design – Permission to HR Manager

- Kasun (Front Office Manager): Providing full privileges to guests', reservations', and payments' data for the front office.

Figure 38: Physical Design – Permission to Front Office Manager

```
-- 4) Front Office Manager - full access to guest and booking related data
CREATE USER Kasun FOR LOGIN FrontOfficeManager;
GRANT ALL ON Guest TO Kasun;
GRANT ALL ON Guest_Phone TO Kasun;
GRANT ALL ON Booking TO Kasun;
GRANT ALL ON Payment TO Kasun;
```

- Ruwan (Housekeeping Supervisor): Providing full privileges to rooms' information and housekeeping task management.

```
-- 5) Housekeeping Supervisor - full access to room and task related data
CREATE USER Ruwan FOR LOGIN HousekeepingSupervisor;
GRANT ALL ON Room TO Ruwan;
GRANT ALL ON Standard_Room TO Ruwan;
GRANT ALL ON Deluxe_Room TO Ruwan;
GRANT ALL ON Suite TO Ruwan;
GRANT ALL ON Housekeeping_Task TO Ruwan;
```

Figure 39: Physical Design – Permission to Housekeeping Supervisor

- Iresha (Finance Manager): Providing full privileges to payments' data and read-only privileges to reservations' data.

Figure 40: Physical Design – Permission to Finance Manager

```
-- 6) Finance Manager - full access to payment records for reporting
CREATE USER Iresha FOR LOGIN FinanceManager;
GRANT ALL ON Payment TO Iresha;
GRANT SELECT ON Booking TO Iresha;
```

- Nadeesha (Receptionist): Providing read-only privileges to rooms' availability and reservations' data.

```
-- 7) Receptionist - view-only access to check room availability
CREATE USER Nadeesha FOR LOGIN Receptionist;
GRANT SELECT ON Room TO Nadeesha;
GRANT SELECT ON Booking TO Nadeesha;
```

Figure 41: Physical Design – Permission to Receptionist

CHAPTER 4: CONCLUSION

Serendib Grand Hotel Database System is an example of a very comprehensive and structured way of dealing with the various difficulties of managing data within a modern hotel. Having

done a very detailed process starting with the analysis, conceptual designing and finally implementing the database, we have come up with a properly structured and normalized database system that reflects the true workings of a single-branch hotel, from room classification to bookings and other various services.

Each of the following structures that were studied in the Data Management 1 unit have been used in designing the database: disjoint specialization of room types, weak entity (managers), multivalued attributes converted to relations, many-to-many relation between services. All the above structures have been included in the normalized schema consisting of fifteen relations. They have been implemented in the SQL Server using the primary keys, foreign keys, CHECK and UNIQUE constraints in order to maintain data integrity. It is clear from the following SELECT, GROUP BY, HAVING and JOIN queries on the above schema that the database will be able to solve the practical problems that would arise in the hotel on a regular basis, along with the seven permissions defined for the roles of the users.

To conclude, the development of this project has provided the Serendib Grand Hotel with a dependable, safe and scalable database framework which will eliminate any data redundancy, inconsistent and invalid data entry and aid in making better decisions faster. In case the hotel decides to scale up its operations in the future, this approach can be adopted to incorporate features like online booking, multiple branches and a separate front desk management system.